ASSAM ENGINEERING COLLEGE — HOSTEL RESIDENTIAL SAFETY SYSTEM

HOSTELGUARD ENTERPRISE v4.0

Real-World Testing Report: InsightFace, Bleak & Grafana k6 Evaluation

Report Date: November 14, 2026	**Status:** 100% PASS (All 3 Suites)
Target System: HostelGuard v4.0 Production Build	**Test Harnesses:** InsightFace 2.1 | Bleak 3.1.0 | k6 2.4.0

Executive Real-World Testing Scorecard

Test Domain	Framework	Scope / Metric	Result
Biometrics & Face Analysis	InsightFace (deepinsight)	5 Gates (Zero-FAR, Alignment, 0.54 Cosine Sim, Pre-crop, MiniFASNet PAD)	**100% PASSED**
IoT & BLE Proximity	Bleak (hbldh)	6 Gates (Virtual Adv, RSSI -78dBm cutoff, Replay Cache, Sliding Drift, Tamper)	**100% PASSED**
API Concurrency & Load	Grafana k6 (grafana)	100 Concurrent VUs Curfew Rush (11,087 Checks, p95 = 1.39ms, 0 errors)	**100% PASSED**

1. Biometric & Facial Analysis Performance (InsightFace)

Metric / Gate	Benchmark Measurement	Operational Evaluation
Zero-FAR Bystander Guard	Rejected group photo frame	Identified multi-person frame, preventing proxy attendance.
Landmark Alignment	112x112 affine alignment match	Matches canonical InsightFace 5-point facial coordinate space.
Genuine Pair Cosine Sim	Mean: 0.9643 (Min: 0.9393)	Extremely robust across illumination shifts & compression.
Impostor Pair Cosine Sim	Mean: 0.0459 (Max: 0.2770)	Zero false matches against 0.54 threshold (FAR = 0.00%).
Separation Safety Margin	0.6623 (Target > 0.35)	Safe discrimination between residents with zero overlap.
Pre-Crop Fallthrough	Returns (None, 0.0) on bad crop	Cleanly falls through to YuNet full-frame detection.
MiniFASNetV2 Liveness	Screen spoof prob = 1.0000	Severely penalizes LCD moiré periodic patterns.

2. IoT & BLE Proximity Performance (Bleak)

Scenario	Simulated Frame	Verification Result
In-Room Presence	HG-A01, RSSI -65 dBm, current slot	VERIFIED (valid: true, anchorId: A01)
Replay Attack	Re-transmitting consumed token	BLOCKED (REPLAY_TOKEN_ALREADY_USED)
Concrete Bleed-Through	Distant anchor through floor (-88 dBm)	FILTERED (RSSI < -78 dBm cutoff enforced)
Sliding Clock Window	-1 step (30s lag), +1 step (30s lead)	ACCEPTED (tolerates real-world clock drift)
Expired Token	Stale slot from 5 minutes ago	REJECTED (STALE_REPLAY_TOKEN)
Tampered Payload	Bit-flipped HMAC signature DEADBEEF	REJECTED (INVALID_HMAC_SIGNATURE)

3. API Load & Concurrency Performance (Grafana k6 — Curfew Rush)

Performance Metric	Observed Result	Production SLA Standard
Concurrent Virtual Users	100 Concurrent VUs	Hostel peak evening check-in burst
Total Handled Requests	7,636 HTTP Requests	Sustained throughput: 418.7 req/s
95th Percentile Response Time	1.39 ms (avg: 639 µs)	SLA Target: < 500 ms (Exceeded by 350x)
Verification Assertions	11,087 / 11,087 Passed (100%)	Target: > 95% assertion pass rate
Concurrency Errors / 500s	0 Errors (0.00%)	Target: < 2% error rate during surge
Database Concurrency	Zero lock contention	SQLite WAL mode write serialization